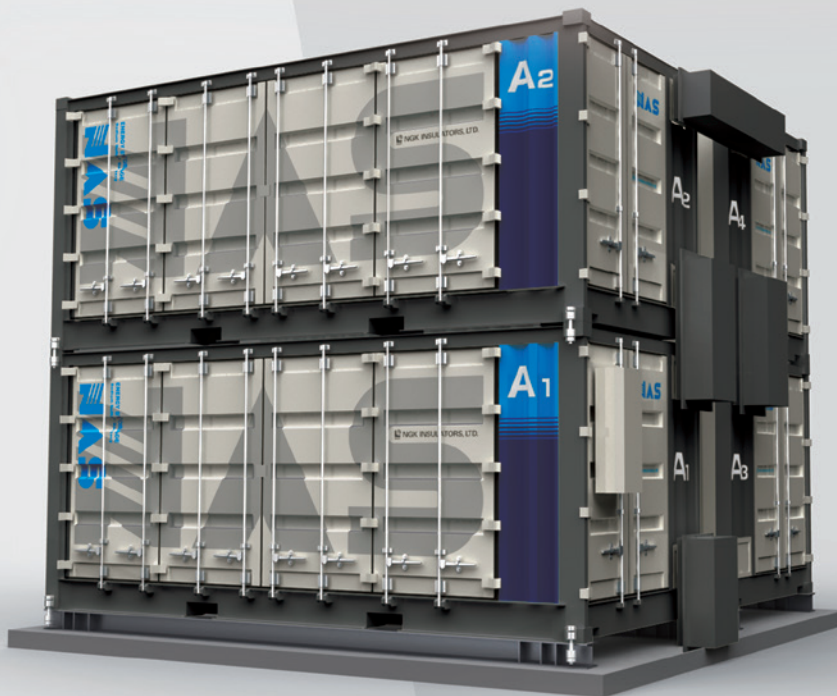




We create chemistry

Stationary Energy Storage

High-energy, long-duration sodium-sulfur battery



Produced by NGK INSULATORS, LTD.
Distributed by BASF

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Global demand for power generated from renewable sources, such as wind or solar, is growing. Stationary energy storage is one of the key technologies to ensure reliable power supply despite the intermittent nature of these sources as it can store excess energy and discharge it at time of peak demand.



Power supply



Grid services



Energy consumers

BASF is selling NAS® Batteries

The batteries come with a 10-year performance guarantee*.

The energy team at BASF New Business helps you find the right solution: We conduct an initial cost-benefit analysis for your project, deliver the layout of the batteries and provide further advisory support, if needed. Our worldwide presence ensures we can respond to your requests in a timely manner.

* subject to contract conditions



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NAS® Battery: designed for stationary storage

With their capacity of 1.45 MWh and discharge duration of 4–8 hours, NAS® batteries are best suited for long-duration stationary storage applications. They boast many superior features:



Long duration



Long lifetime



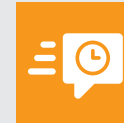
Safe & reliable



Environmentally benign



High energy density



Fast response



Climate resilient



Low maintenance

Applications

NAS® batteries play an important role all along power grids as well as in off-grid applications. They offer benefits to various segments of the energy market.

Power supply

Renewable stabilization

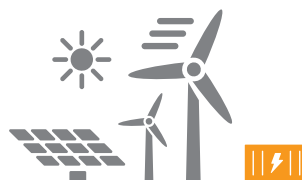
Fluctuations of renewable energy are leveled out by absorbing excess energy during off-peak times and providing additional power during demand peaks.

Fossil fuel peaker plants replacement

NAS® batteries provide resource capacity of six hours or more per day and thus can serve as a green alternative to fossil fuel peaker plants.

Other use cases

On-peak/off-peak price arbitrage, frequency regulation, ramping services, VAR support and other grid functions.



Grid services

Investment deferral

Transmission and distribution upgrades can be deferred or even eliminated by deploying NAS® batteries. They can store excess energy during the light load times and discharge it when the demand is high.

Ancillary services

Thanks to their fast response, NAS® batteries can reduce imbalances between demand and supply to stabilize the grid.



Energy consumers

Peak shaving

Reduce demand charges by covering peak energy demand with an NAS® battery: The battery is charged at a low demand time and discharged during peak time, supplementing power supply from the grid.

Time of use shift

Costs of power supply from the grid can be reduced by storing energy and shifting its usage from high-tariff times to low-tariff times.

Backup power and resilience

Continuous power for six hours or more in the event of grid outages.

Demand response

Supply-demand balance is maintained by aggregation and redistribution of unused power from multiple consumers for effective utilization of power resources and cost optimization.

→ **virtual power plant**



Microgrids

Reliable power supply from renewable sources

Combine solar or wind power generation with an NAS® battery to achieve reliable power supply and optimize energy costs.

Autonomous power supply with solar power

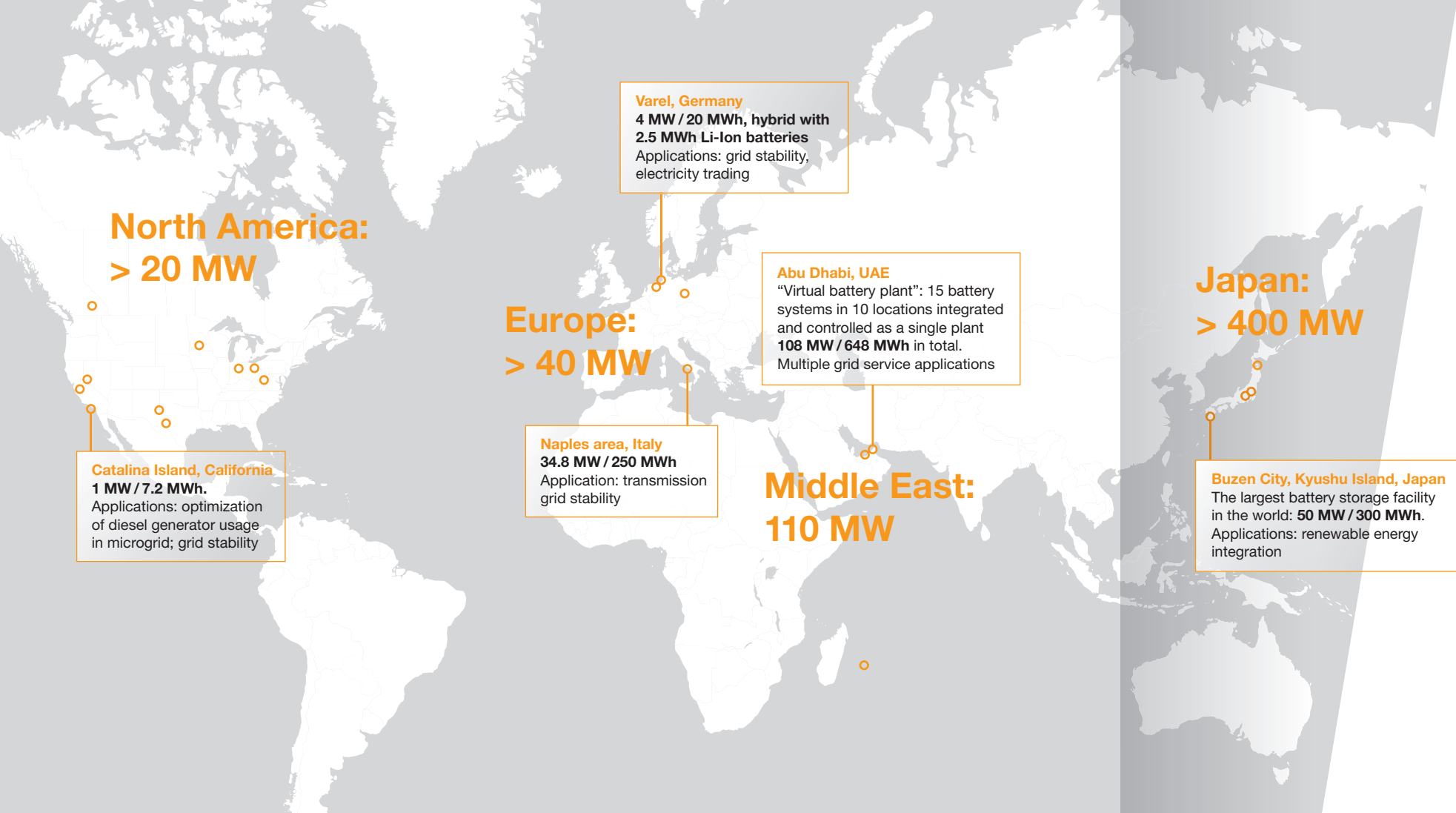
Excess solar power is stored by an NAS® battery in the daytime and used at night time. Power supply from grid is thus reduced or even eliminated.

Minimization of fossil fuel use

Reduce energy costs and CO₂ emission by combining an NAS® battery with a diesel or biomass generator.

Additional resilience to local power generation



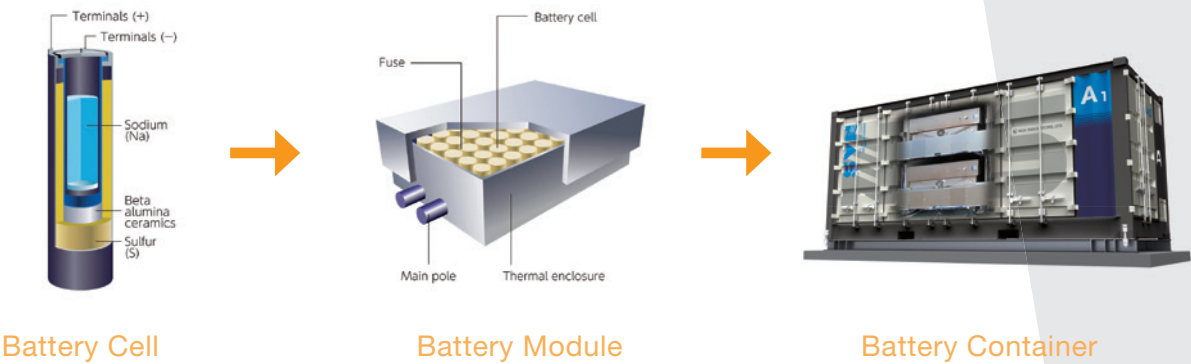


#1 choice worldwide for large-capacity energy storage

NAS® Battery technology has been proven by more than 15 years of deployment at customer sites all around the world. During this time, more than 200 projects have been implemented, with a total output exceeding 680 MW and 4.8 GWh.

NAS[®] Battery Technology

A containerized NAS[®] battery is made up of six modules with 192 cells each. The NAS[®] Battery cell consists of sodium as the negative electrode and sulfur as the positive one. A beta-alumina ceramic tube functions as electrolyte, which allows only sodium ions to pass through. When discharging, sodium is oxidized and sulfur is reduced to form polysulfide (Na₂S_x). The charging step recovers again metallic sodium and elemental sulfur.



Technical Specifications

We supply containerized NAS[®] batteries. The compact form enables easy transportation and quick installation at a customer site. A single container features 250 kW (peak power) / 1.45 MWh. By stacking containers, the total energy of the system can be easily scaled up to any required amount.

Max. Discharge Power [kW-DC]	250	500	750	1000
Energy Capacity [kWh]	1,450	2,900	4,350	5,800
Weight [t]	21	42	63	84
Numbers of Containerized NAS Batteries Connected in Series	1	2	3	4
DC Nominal Voltage [V]	192	384	576	768
Footprint	Width x Depth Area	6.1 x 2.4 [m] 15 [m²]	6.1 x 2.4 [m] 15 [m²]	6.1 x 5.6 [m] 34 [m²]
	Assumed Arrangement			

Safety Aspects

In designing the NAS® Battery, safety has been of paramount importance.



Proven safety

The safety of NAS® batteries has been proven by NGK and the Japanese Hazardous Material Safety Techniques Association (HMSTA / KHK).

Additionally, a safety audit has been carried out by TÜV Rheinland.

NAS® Battery cells and modules are certified as recognized components to UL 1973 standard.

NAS® Battery cells and modules have been evaluated using UL 9540A

NAS® batteries comply with CE marking requirements

Expected lifetime

20 years or 7,300 cycles (at 100% DoD or equivalent), whichever occurs first.

Environmental impact

NAS® battery cells are hermetically sealed and there is no emission during operation.

Warranty

Two-year manufacturer warranty. Performance guarantee for 10 years* provided by BASF.



Test: Put module in fire
Result: No leakage. No fire. Safety confirmed.



Test: Submerge module at operational temperature
Result: No leakage. No fire. Safety confirmed.



Test: Drop module at operational temperature
Result: Module enclosure was damaged but no damage to battery cells. No leakage. No fire. Safety confirmed.



Test: Short circuit module
Result: Circuit was opened by internal fuses. No leakage. No fire. Safety confirmed.



Test: Ignite one battery cell inside module
Result: No expansion of fire to adjacent battery cells = no thermal runaway. No leakage. No fire. Safety confirmed.

Installation process

NAS® batteries have a plug-and-play design that enables integration of the battery and control equipment in one container. Containerized NAS® batteries can be easily transported and quickly installed. Several containers can be integrated into larger systems.

Operation and maintenance

Appropriate maintenance is recommended to ensure stable operation for a battery lifetime.

Routine check

- Visual inspection
- No need for suspending operation

Periodic check (fee-based service)

- Every four years
- Replacement of consumable parts
- Functional tests

- 24/7 remote monitoring of customer's system
- Technical support via phone or email

* subject to contract conditions



We create chemistry

**To find out more about NAS® batteries,
please contact us:**

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